

Extended Essay

Physics

Topic: The physical merits of a mix of thorium and plutonium fuel salts as a substitute to uranium fuel salts in fluoride Molten Salt Reactors

Research Question: To what extent do the optimised neutronics of Pu-239 and Th-232 fluoride salts make them a substitute for uranium fluoride fuel salts in the Molten Salt Reactor Experiment?

Examination Session

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1. Introduction:

This Extended Essay will address the research question: To what extent do the optimised neutronics of Pu-239 and Th-232 fluoride salts make them a substitute for uranium fluoride fuel salts in the Molten Salt Reactor Experiment?

Since before I began this journey, I have aspired to be a nuclear energy engineer, because to me, it seems to be an exceptional way to combat man-made climate change. Technologies which promise to cheaply, reliably, and cleanly produce energy at scales are essential to reaching net zero emissions (IEA, 2024). However, the dependence on uranium remains a bottleneck in nuclear energy's potential, especially in countries with low uranium reserves. Technologies that promise superior neutron economy, such as MSRs (Molten Salt Reactors) and reactors using thorium-plutonium fuel cycles, are critical to the global viability of nuclear energy (Vlasov, 2023).

A Molten Salt Reactor is defined as a reactor which uses liquid salt as fuel and coolant. A fuel salt is defined as the part of the liquid salt which contains the radionuclides used for the reactions (IAEA, 2016). While learning about nuclear energy, and its potential for decarbonisation, I began learning about molten salt reactors, and in particular the 1965 Molten Salt Reactor Experiment (MSRE), an experiment aiming to show the viability of the Molten Salt Reactor as the next energy source (ORNL, 1972). I had been reflecting on methods to make nuclear energy renewable, from a physics perspective, and as a result, I found the EE to be an ideal way to pursue this fascinating and globally relevant topic.

To model the fuel mix of the molten salt reactor, with the conversion from use of uranium to thorium and plutonium, a simulating script based on the MSRE, simulation software will be used.

DAGMC (Svalinn, 2023) and OpenMC (Romano and Forget, 2013), will simulate the reactor geometry and neutronics, respectively. A simulating script written by Copenhagen Atomics will be utilised, due to the infeasibility of using the real MSRE.

In both simulators, and real-world trials, fission experiments have significant uncertainties, due to extreme conditions, such as high temperature and radiation (ORNL, 1972). Since the MSRE ended in 1969, many sources will be primary sources which were produced decades ago (ORNL, 1972). This essay will focus on neutronics, such as the cross sections of different nuclear reactions.

2. Hypothesis:

The substitution of an MSRE based reactor model, using plutonium and thorium salts instead of uranium, due to the fissile properties of plutonium-239 and the fertile properties of thorium-232 make it a more powerful fuel mix, yielding on average 10-15% higher k-effectives.

K-effective is defined as the ratio of the rate of total neutrons produced to the rate of total neutrons lost and is used to measure the ability of a reactor to increase its total neutron population (Oka and Suzuki, 2013).

Firstly, the difference in fissile and cross-sectional properties between U-235 and Pu-239 (Duderstadt and Hamilton, 1976), and Pu-239 having a higher neutron multiplication factor than U-235, make a plutonium-thorium mix more reliable for sustaining a chain reaction (IAEA, 2014).

Secondly, the difference in fertile material, between Th-232 and U-238 results in major differences in k-effective (Duderstadt and Hamilton, 1976), due to the difference in potency between their eventual beta decay daughters after neutron capture: U-233 and Pu-239 (Rehn, 2014). Fertility is the ability to produce fissile material through neutron capture (Duderstadt and Hamilton, 1976). This investigation will assume the percentage of radioisotopes other than Th-232 and Pu-239 in the thorium and plutonium fuel salt is negligible, and therefore keep these percentages in the salt constant (Knudsen, 2022).

3. Theory:

The Molten Salt Reactor Experiment was simulated, due to its significance in the history of MSRs. MSRE simulators were more user-friendly and easier to access, compared to other simulations, which either required technical skills outside of the scope of a Physics Extended Essay or were not accessible to an IB student.

Enrichment is the portion of a fissionable sample of radionuclides which is a fissile radioisotope (WNA, 2022). The MSRE's test reactor utilised a uranium fuel salt, with a 0.33 enrichment (ORNL, 1972), which is much higher than the enrichment of the fuel for traditional light water reactors (WNA, 2022). Since the deposits of plutonium produced by current nuclear reactors and thorium as byproducts of mining would be used, there would likely be natural thorium (mainly Th-232) and "natural" plutonium, such as in other molten fluoride reactors (IAEA, 2016). However, due to issues with obtaining cross-sectional data for certain radionuclides, this investigation assumes the plutonium is entirely plutonium-239.

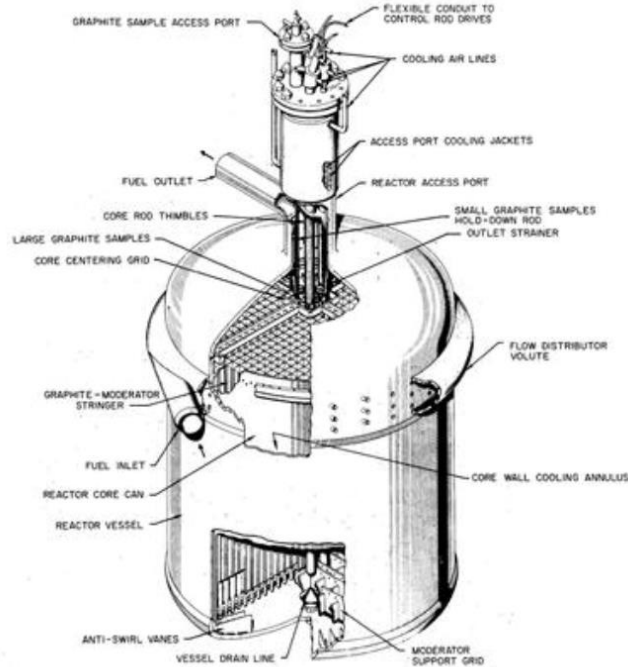


Figure 1: The fully labelled simplified reactor schematic of the MSRE, showing graphite moderating stringers, a base of horizontal graphite bars and a moderated liquid fuel (ORNL, 1972)

This model, seen in Figure 1, is an accurate approximation of the MSRE design, emulating the key original design choices of the MSRE, including heat generated in the core. It emulates the heat transfer from the fuel salt to the coolant salt (in the primary heat exchanger), and a fuel pump circulating liquid fuel (ORNL, 1972). Due to the focus on neutronic properties, other parts less relevant to neutronics will not be a focus. The graphite moderator of the MSRE and coolant salts in the vessel are emulated in the experiment's code. This experiment has computational limitations, because a maximum of 20,000 particles were rendered for each trial, potentially skewing results. Higher particle numbers were not viable, as I do not have access to hardware which can render more particles.

Due to **OpenMC's tendency** to have **its most accurate value** for **k-effective** be the **first value**, and the exact numerical values being put into Excel for data processing, there is not a significant random error from measurement. The only error is the systematic error from the conditions of the simulation. When **cross-referencing** other **studies** which **modelled** the **Molten Salt Reactor Experiment** in OpenMC, **similar values** were reached. For example, the study conducted by the University of California Berkeley, by (Fratoni et al, 2020), **reached** the **same value**, of **0.998** for the **k-effective** of the **MSRE** running at **0.33 enrichment**, reflecting technological limitations in reactor simulation.

The main measurement of reactor stability and reactivity will be k-effective, also known as neutron multiplication factor, as below (Oka and Suzuki, 2013). The goal of these fuel optimisations is a slightly supercritical reactor where k-effective is roughly 1.02-1.03 (Fratoni et al, 2020), which is safely self-sufficient.

$$k_{effective} = \frac{\text{Neutrons Produced from Fission}}{\text{Neutrons Lost from Capture} + \text{Neutrons lost from Leakage}}$$

It is worth noting, that **the instantaneous neutron population** is proportional to **instantaneous production of thermal energy**, and as the **k-effective is the neutron multiplication**, the k-effective also **measures the ability for a reactor to increase its power** (Oka and Suzuki, 2013). Seeing the formula, we understand that when k-effective is 1, it is critical where neutrons lost equal neutrons gained, and when it exceeds 1 it is supercritical and able to increase its power (Oka and Suzuki, 2013).

It must be remembered that neutrons lost includes leakage from the reactor core, and neutron loss from capture (by all nuclides). Leakage factor is the decimal ratio of the neutrons leaked from the reactor core to the total number of neutrons (Duderstadt and Hamilton, 1976).

For fuel effectiveness, in later cases of this experiment, the cross sections of the nuclides will be considered. Cross section is the surface area where a neutron can collide with a nucleus (ideally for capture if it's a fissionable nuclide), so it can fission or decay into fissile fuel (Duderstadt and Hamilton, 1976). As such, for this essay, the cross sections will be of major focus due to their subsequent effects on nuclear reactions (Oka and Suzuki, 2013). There are two types of cross sections which will be considered:

Neutron Capture:

The incident neutron is captured by the nucleus, where the resultant nucleus is excited and undergoes beta decay (Duderstadt and Hamilton, 1976). Fertile radioisotopes, such as U-238 and Th-232, can beta decay into fissile isotopes Pu-239 and U-233 respectively, supporting the baseline fission chain reaction (Duderstadt and Hamilton, 1976).

Considering Thermal and Fast Neutrons:

Since the conditions in cross section experiments are different to the dense neutronic flux of a thermal molten salt reactor like the MSRE, cross-sectional data must only be used for analysis relative to cross-sectional data of other nuclides from the same database, to control for systematic errors (IAEA, 2014). Thermal neutrons (neutrons where the most probable neutron energy of the Maxwell-Boltzmann distribution is 0.025 eV at 290K) lead to a more consistent chain reaction (Duderstadt and Hamilton, 1976). This is because fast neutrons have much higher cross sections for nuclear reactions, which are detrimental to the fission chain reaction (IAEA, 2014). These include scattering (where the neutron is deflected and loses kinetic energy) and the neutron absorption of non-fissionable nuclides, such as the zirconium (IAEA, 2014). Furthermore, in preliminary trials testing neutron energies, thermal neutrons were seen having much lower leakage factors due to lower penetration power and yielding higher k-effectives.

These cross sections will be used to show relationships, where the neutrons which collide are more or less likely to fission and collide elastically or inelastically, with certain radionuclides at certain energies.

4. Variables:

4.1 Independent Variable:	How it will be changed (In a simulation the data is controlled in the code), How it is significant
Plutonium-239 Concentration Relative to the Concentration of Total Radionuclide Salts (No Units) (0-1)	The relative Plutonium-239 concentration is a significant independent variable because it directly affects the concentration of fissile material in the experiment and affects the resulting k-effective. The line between subcriticality, criticality and supercriticality is mostly affected by the concentration of fissile nuclides, in this case plutonium. The proportion of plutonium, the only nuclide in this fuel salt with adequate neutron multiplication to sustain the chain reaction (Duderstadt and Hamilton, 1976), will be the independent variable. Values will be set in the code.
Thorium-232 Concentration Relative to the Concentration of Total Radionuclide Salts (No Units) (0-1)	The relative thorium concentration is a significant independent variable. It directly affects the amount of fertile material in the experiment, helping replenish the fissile material in the reactor (Vlasov, 2023). Additionally, it increases the k-effective, stability and sustainability of the reaction. Values will be set in the code.
4.2 Control Variable:	How is it significant and controlled?
Coolant Concentration	The relative concentration of coolant salts which lack radioisotopes, such as LiF and ZrF_4 is consistent in this simulation. Values are set in code.
Reactor Moderator	The choice of Graphite as the reactor moderator will not be changed in this experiment, to emulate the MSRE with maximum accuracy. The moderator code will not be changed.
Use of Control Rods	Control rods with neutron poison (nuclides designed to rapidly end a fission chain reaction in emergency situations) will not be used in this experiment (Tucker, 2019).
Reactor Shape and Material	The geometry of the reactor and structure will remain unchanged and function as a control variable. The containment vessel will have the same Hastelloy (a corrosion resistant nickel alloy used in the MSRE) vessel, so the leakage factor is only changed by changes in fuel salt composition. (ORNL, 1972)
Particle Quantity	The quantity of particles rendered in the simulation (20,000), will be unchanged. There will be multiple trials of the experiment (20 batches) to ensure a precise result without significant random uncertainty.

Table 1: A table containing the independent and control variables of this Extended Essay, including how they will be controlled or modified and how they are significant to this investigation.

4.3 Dependent Variable:	How it will be affected/measured (In a simulation data is found in tallies)
k-effective (No Units) (0-1.5)	It is the factor at which the neutron population changes in each generation, directly affected by the concentration of fissile and fertile radioisotopes in the fuel salt which can support the reaction and maintain neutron population. The analog k-effective value is used. The analog k-effective value measures the number of fission sites in a cycle to determine the change in neutron population (Romano and Forget, 2013). k-effective also indicates the speed at which a fission chain reaction can accelerate, and since the number of fission reactions and reactor power are proportional, the rate at which power can be increased (Oka and Suzuki, 2013)
Leakage Factor (No Units) (0-1)	The decimal percentage of neutrons leaked in each generation of the reaction, affected by the relative concentration of the radioisotopes in the fuel salt (Duderstadt and Hamilton, 1976). Measured by the quantity of neutrons in the containment vessel before and after a cycle (Romano and Forget, 2013).

Table 2: A table containing the dependent variables of this Extended Essay, including how they are affected and measured.

5. Methodology:

5.1 Equipment:

A powerful computer running a Linux distribution based on Arch, Fedora or Debian (If Linux is not installed a virtual machine would work) was used. This experiment was conducted with a PC with an RTX 2060 installed. The following procedure was executed on Linux Fedora, and it must be considered that WSL (Windows Subsystem for Linux) would require additional steps for configuration. Please note a device with the anaconda package manager installed is required.

5.2 Procedure:

1. OpenMC was installed (Romano and Forget, 2013)
2. In the Linux Terminal, the following commands were used to install OpenMC in a container.

```
conda config --add channels conda-forge
```

```
conda create -n openmc-env
```

```
conda activate openmc-env
```

```
conda install mamba
```

```
mamba search openmc
```

```
mamba install openmc (Romano and Forget, n.d.)
```

3. A folder labeled OpenMC was created in the home directory, where a file called "MSRE.py" was made, containing the code of this website (Knudsen, 2022):

https://github.com/openmsr/msre_docker/blob/main/MSRE.ipynb

4. The ENDF/B-VIII.0 library from <https://openmc.org/official-data-libraries/> was downloaded and extracted into the OpenMC folder of the home directory (Romano and Forget, n.d.).
5. DAGMC was downloaded, with the library extracted into the OpenMC folder, and renamed to DAGMC. The folder “CrossSections” was renamed.
6. The following header was added to the OpenMC input file so the program can access cross section values.

```
import openmc

import os

os.environ['OPENMC_CROSS_SECTIONS']=

"/home/USERNAME/OpenMC/CrossSections/endlb-viii.0-hdf5/cross_sections.xml"
```

7. To emulate the neutronics of the MSRE, the energy_distribution variable for the neutron source was changed to *openmc.stats.Maxwell(0.025)* which approximated the neutron energy curve as a Maxwell-Boltzmann energy distribution where 0.025 eV was the most likely energy (Duderstadt and Hamilton, 1976), (Romano and Forget, n.d.)
8. The fuel salt composition was replaced with this molar composition:

```
molar_comp={'LiF':0.65,'BeF2':0.291, 'ZrF4':0.05, 'UF4':0.009}
```

This composition was more accurate to the original MSRE conditions (ORNL, 1972)

9. To run the simulation the following commands were run in the terminal:

```
cd OpenMC

python MSRE.py
```

6. Results:

6.1 Case 1: Base Case of the MSRE: The relationship between Uranium

Enrichment and k-effective:

Bat./Gen.	k	Average k
=====	=====	=====
1/1	0.98803	
2/1	0.56488	
3/1	0.80336	
4/1	0.88642	
5/1	0.90615	
6/1	0.90641	
7/1	0.90217	0.90429 +/- 0.00212
8/1	0.92657	0.91172 +/- 0.00753
9/1	0.90747	0.91066 +/- 0.00543
10/1	0.91862	0.91225 +/- 0.00449
11/1	0.91100	0.91204 +/- 0.00368
12/1	0.92416	0.91377 +/- 0.00356
13/1	0.93993	0.91704 +/- 0.00449
14/1	0.93201	0.91870 +/- 0.00430
15/1	0.93150	0.91998 +/- 0.00405
16/1	0.92194	0.92016 +/- 0.00367
17/1	0.93474	0.92138 +/- 0.00356
18/1	0.93165	0.92217 +/- 0.00337
19/1	0.94780	0.92400 +/- 0.00362
20/1	0.92142	0.92383 +/- 0.00337

```

k-effective (Collision)      = 0.92377 +/- 0.00301
k-effective (Track-length)  = 0.92383 +/- 0.00337
k-effective (Absorption)    = 0.92363 +/- 0.00196
Combined k-effective        = 0.92362 +/- 0.00210
Leakage Fraction            = 0.32808 +/- 0.00126

```

Figure 2: The data from the base case of the simulation, with 20 trials and a final dataset at the end with values for the resultant k-effectives. Analog (absorption) k-effective values will be used due to their simplicity and accuracy (Linux Terminal, 2023)

It is worth noting, before proceeding with the results, that each value will have 20 trials, but only the first trial will be shown. Through cross comparison with other literature on the neutronics of the MSRE, only the first k-effective value was accurate, due to OpenMC's tendency to underestimate subsequent k-effective values.

The **literature value** for **k-effective** at 0.33 enrichment is **0.988**, the first value seen in Figure 2 (Fratoni et al, 2020). Notably, the values after were all far from this literature value. Averages in the simulation software are not calculated until the seventh trial, due to high initial variances between values.

The value for average k-effective (0.91) would mean that the reaction's neutron population is rapidly declining, and that a fission chain reaction will not be sustained. The leakage fraction is at 0.33, within the range of this investigation, which is 0.3-0.4.

Previous tests with 0% enrichment for both variables, using fast neutrons and standard neutronic configurations, netted k-effective with an order of magnitude in the thousandths. This reflects the real world, where no matter how much fertile U-238 or Th-232 are piled, there is no reaction.

Thorium is not fissile, but rather fertile, meaning it can decay and fission, but its neutron multiplication factors are not adequate to sustain the reaction, unlike enriched uranium (IAEA, 2014), (Duderstadt and Hamilton, 1976). A pile of Th-232 without any fissile isotopes to support the reaction would not fission. This also applies to U-238, which is only fertile, thus depleted uranium (Uranium with lower concentration of fissile uranium isotopes than natural uranium), is incapable of sustaining a fission reaction (IAEA, 2014). This necessitates fissile material well above natural concentrations, requiring the enrichment of uranium or a modified mix of plutonium and thorium.

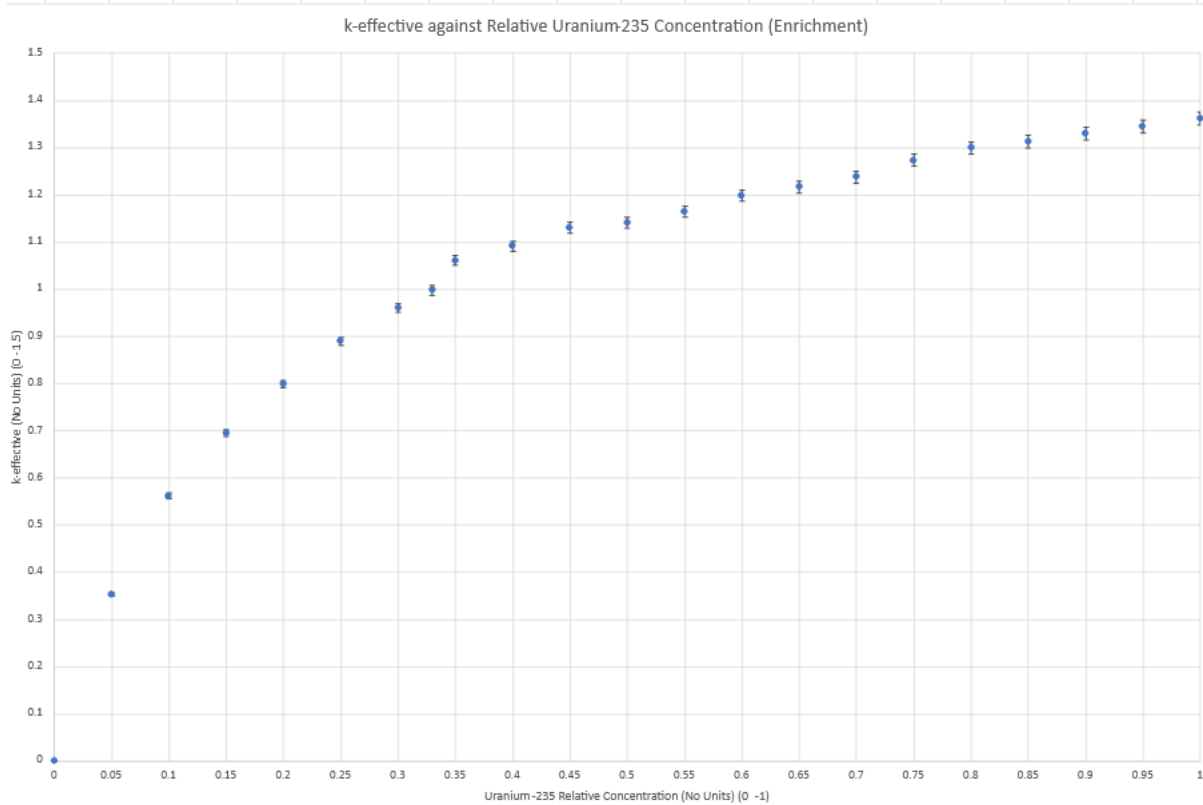


Figure 3: Showing the relationship between enrichment and k -effective in a U-235-U-238 based MSRE fission simulation with the k -effective shown as a function of the enrichment value, with the additional y value at 0.33 corresponding to default enrichment. Error bars correspond to software generated values for random uncertainty. (Excel, 2024)

Although uneven, a trendline with a derivative decreasing with an increase in the value of enrichment can be seen. This makes logical sense, because every other variable is controlled. Eventually, the limiting variable will not be the concentration of fissile isotopes, but rather, variables which will not be changed in this experiment, because this reactor was designed for uranium fuel at 0.33 enrichment (ORNL, 1972).

Leakage factors stayed at around 0.35 throughout the simulation, meaning that it is more dependent on the control variables, such as the nickel alloy shell and lithium/zirconium salts. If there are more neutrons, a proportional quantity of these neutrons will leak.

The decreasing derivative means that even without control rods inserted, the maximum possible k-effective, even in the impossible case of enrichment=1, is 1.4.

The U-235 plays the role of the fissile material. The formula below shows the equation of U-235 fission, obeying the conservation of baryon number and conservation of energy. It can be seen for every neutron used for fission, multiple fission neutrons are produced (Koelsch, 2016):

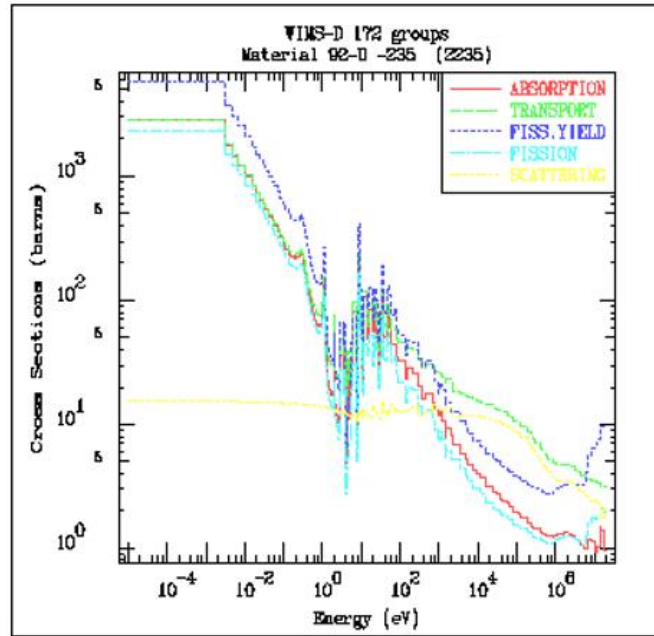


Figure 4: U-235 Fission Cross Section, with the fission cross section highest in the thermal range (IAEA, 2014)

Barns, as seen in Figure 4, are an SI unit 10^{-28}m^2 which measures the cross-sectional area where a neutron can hit a nucleus and undergo a given nuclear reaction (Duderstadt and Hamilton, 1976). The higher the cross section is in barns, the higher probability of the given nuclear interaction. In Figure 4, one can see the fission cross section is highest at the thermal energy range.

With the 3 neutrons, the average kinetic energy of the product neutrons is 2 MeV (Koelsch, 2016). These are fast neutrons (Duderstadt and Hamilton, 1976), and in the reactor's environment, they must be slowed down into thermal neutrons for a more stable reaction with a lower leakage factor. This would lead to a safer and more efficient reactor. These three neutrons sustain the fission reaction, by supplementing the neutrons needed for U-238 reactions. U-238 is a fertile material, meaning it cannot sustain a fission chain reaction. However, U-238 can turn into Pu-239 through the following decay reactions (Chidambaram and Ganguly, 1996):

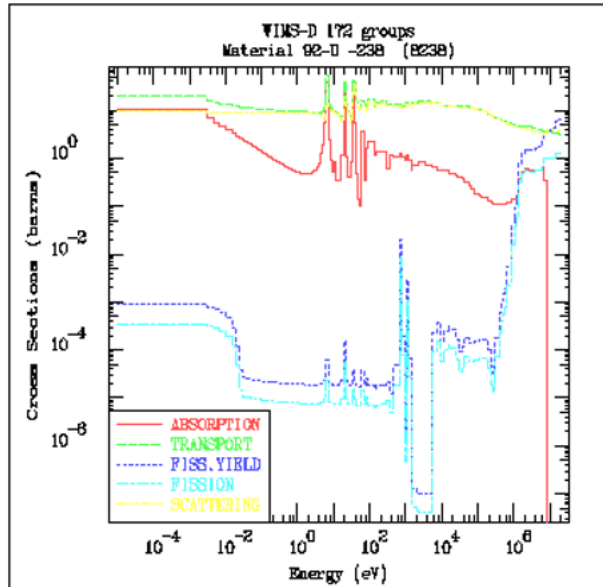
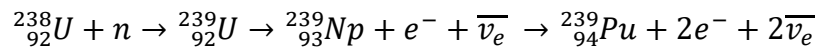


Figure 5: U-238 Fission cross section, showing that the fission cross section is orders of magnitude lower than for the fissile isotopes, especially in thermal ranges (IAEA, 2014)

Comparing the cross-section graphs of the U-235 and U-238 seen in Figure 4 and 5, respectively, the U-238 has a much lower fission cross section, which is the reason its main purpose is to supplement the U-235.

For this reason, the k-effective steadily increases with the increase in U-235 concentration, because the more enriched the uranium is with U-235, the more particles there are which can gradually increase the neutron population of the reactor, resulting in a higher k-effective.

Overall, the U-235-U-238 fuel mix remains a golden standard in nuclear reactors. However, the Th-232-Pu-239 still achieved consistently higher k-effectives than the U-235-U-238, which shows that it is a more potent and energy dense fuel salt.

6.2 Th-232-Pu-239 Mix: Relationship between Pu-239 Concentration and k-effective:

The UF_4 must be replaced with a blend of ThF_4 and PuF_4 with a 2:1 ratio to emulate Uranium at 0.33 enrichment (where there is twice as much fertile U-238 as fissile U-235). This is achieved with the code below:

To add the appropriate nuclides, the following section must be removed from the program:

```
salt.add_nuclide('U-235',enrichment*molar_comp['UF4']*1/5,'ao')
```

```
salt.add_nuclide('U-238',(1-enrichment)*molar_comp['UF4']*1/5,'ao')
```

It must be replaced with the code below:

```
salt.add_nuclide('Th-232',(1-enrichment)*molar_comp['UF4']*1/5,'ao')
```

```
salt.add_nuclide('Pu-239',enrichment*molar_comp['UF4']*1/5,'ao')
```

Where $\text{molar_comp}[\text{'UF4'}]$ is the concentration of fissionable salts and enrichment is the proportion of the radionuclides which are fissile (in the base case U-235 but in this case Pu-239). The fertile radionuclides comprise the decimal percentage (1-enrichment) of the radioactive salts.

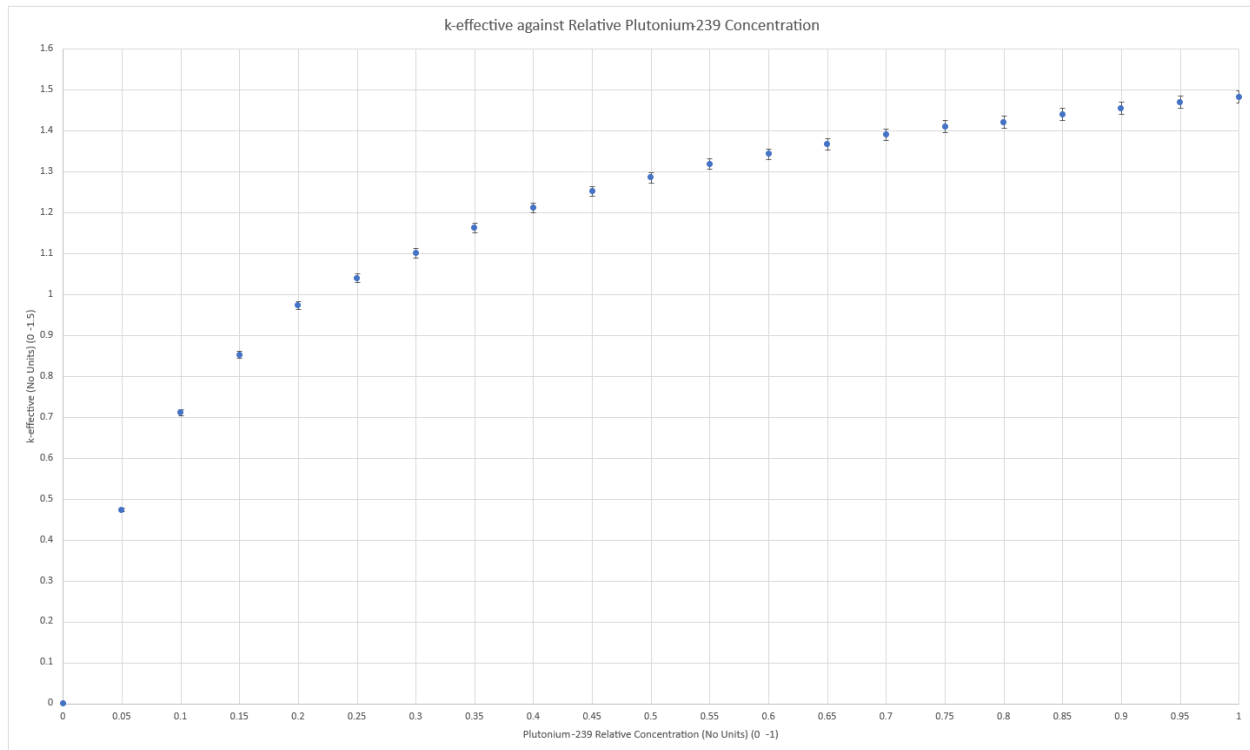


Figure 6: k -effective as a function of the decimal percentage of the fissionable salts which contain fissile Pu-239 fluoride salts, with the remaining fissionable salt being fertile Th-232 fluoride salts. Error bars correspond to software generated values for random uncertainty (Excel, 2024)

Similar to the U-235-U-238 graph, this graph also has a derivative which decreases as the independent variable increases. However, the curve is considerably higher because the Pu-239 is a much stronger fissile nuclide, and Th-232 is more fertile. This means Th-232 can assist the chain reaction more than U-238, which has a higher chance of unproductive nuclear reactions which do not breed more neutrons (IAEA, 2014).

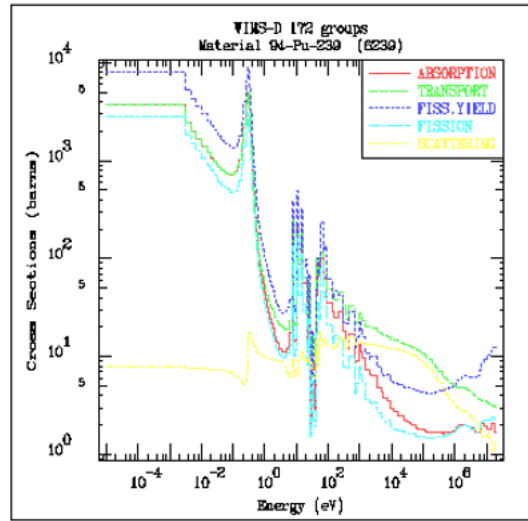
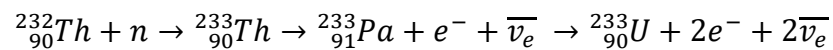


Figure 7: Pu-239's Fission Cross Section, which is much higher than for U-235, also peaking at thermal energies (IAEA, 2014)

Firstly, the fission cross sections of Pu-239 are consistently larger than for U-235, meaning that there is a higher fission probability in a plutonium salt or lattice (IAEA, 2014). The graph is also more shifted to the left, with higher cross sections at lower neutron energies, making Pu-239 more likely to fission than U-235 when impacted by thermal neutrons (IAEA, 2014). This is the reason that Pu-239 as a fissile fuel can yield higher k-effectives, because it is able to fission more readily than U-235.

Th-232 is also a much more effective fertile radionuclide than U-238, not because of a necessarily higher cross section, but due to its fissile daughter once it absorbs a neutron (Chidambaram and Ganguly, 1996):



U-233 may have a similar cross section to the other fissile uranium nuclides, such as U-235 and a worse cross section than the daughter of U-238 (Pu-239), but the biggest benefit for U-233 is

the ability to consistently produce more neutrons at lower neutron energies (Rehn, 2014). This means that it can breed more of itself (U-233) from the Th-232 nuclides.

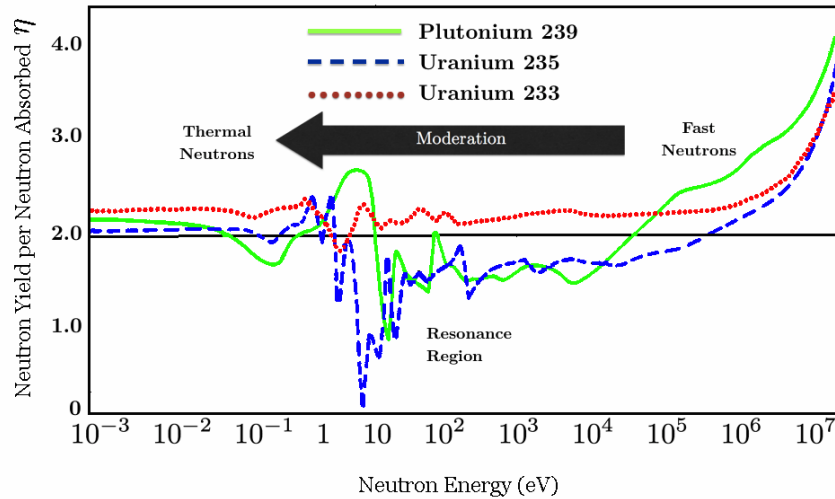


Figure 8: A graph of the neutron multiplication per absorption of U-233, U-235, and Pu-239 (Rehn, 2014)

As seen in Figure 8, in the thermal neutronic range that exists in 290 K (0.025 eV being the most common neutron energy) (Duderstadt and Hamilton, 1976). U-233 has a considerably higher neutron yield than other fissile nuclides. Although thorium itself would not be able to have a chain reaction, even when compared to a natural uranium fuel system, the potential for its daughter U-233 in thermal breeding makes it more effective in neutron multiplication when paired with a fissile nuclide (Rehn, 2014).

Th-232, with some Pu-239 to sustain the chain reaction, outperforms with a higher k -effective, due to the better performance of Pu-239 than U-235 in lower neutron energies. Additionally, thorium can breed U-233 instead of Pu-239, which is a better fissile fuel than both Pu-239 and U-235 at these energies due to its ability to produce more neutrons and to sustain a chain reaction (Rehn, 2014).

Overall, a mix of thorium and plutonium, as an alternative to traditional uranium is a viable alternative to uranium which can complement the current fuel cycle. Plutonium is a common byproduct of modern light water reactors due to neutron absorption of U-238, which beta decays into Pu-239, whereas thorium is a common byproduct of mining rare earth and heavy metals (Chidambaram and Ganguly, 1996). Due to the abundance of Th-232 and Pu-239's effectiveness as a fissile radionuclide, its usage makes it a more sustainable alternative to the uranium fuel cycle (IAEA, 2014).

Pu-239 is a stronger fissile fuel than U-235, with a consistently higher neutron multiplication factor and fission cross section, while Th-232 is a much more effective fertile material because its daughter nuclide U-233 has a much higher multiplication factor than U-238's daughter nuclide Pu-239 in thermal neutronic conditions (Rehn, 2014).

7. Evaluation:

Strengths:

Using a mix of Th-232 and Pu-239, which uses the Pu-239 to sustain breeding U-233 from Th-232, the fuel mix effectively captures the benefits in the neutron economy of breeder reactors and light water reactors. The k-effective is still stable, and simple to control. If the reaction gets too powerful, control rods are inserted. Additionally, the reactor effectively "eats its own waste", because the reaction of Th-232 absorbing a neutron and producing U-233 helps create a more consistent source of fissile material, even when the Pu-239 initially added starts to deplete. In the real world, this means a much more efficient neutron economy.

Limitations:

There is only so much an EE can consider on nuclear reactor physics, and this section will cover the limitations of this research. Although a similar Berkeley study also obtained 0.988, this is still 2-3% off the real-world experimental value of 1.02-1.03 for the k-effective of uranium fuel salts in the MSRE at 0.33 enrichment (Fratoni et al, 2020). Cross Section Data from ENDF-VIII.0 was selected due to its recency and compatibility with OpenMC and was used with the original simulation which the code of this simulation is based on. However, JENDL is more accurate for neutronics simulations (Fratoni et al, 2020). Only the k-effective value from each trial was used because of OpenMC's underestimation of k-effective values after the first trial. This method of interpreting OpenMC output data, although increasing random error, was also used in the Berkeley Study to obtain similar k-effective values (Fratoni et al, 2020).

The critical mass of plutonium was not considered, and in the real world there be concerns the fuel salt could self-fission, and this issue would be intensified if it is not Pu-239 but rather plutonium as a byproduct of reactors, which contain Pu-240 and Pu-241 (WNA, 2023). The use of a simulation with a mid-range PC means that a maximum of 20,000 particles could be rendered, which cannot fully represent the octillions of particles which would exist in a real molten salt reactor.

8. Extension:

To extend this research, there would need to be more emphasis on the "natural" plutonium, which was only Pu-239 in this. The simulation's cross-sectional data was not reliable enough to have elemental plutonium in the simulations, due to the probability of self-fission of Pu-240 and Pu-241, which also happen to have very low critical masses. For an accurate k-effective value for elemental plutonium, there would need to be more emphasis on the Pu-238, Pu-240 and Pu-241 nuclides. Though it may be argued that the reactor grade plutonium may be "enriched" to only compose of Pu-239, this would not account for the mass differences. In uranium enrichment, a gas centrifuge separates radioisotopes with the approximately 1.27% mass difference of uranium, where U-235 can be concentrated in parts of the centrifuge (WNA, 2022). However, Pu-239 and Pu-240 have a much lower mass difference, and the corrosive properties of plutonium salts make this experimentation much more complicated (WNA, 2023). Further opportunities lie in investigating corrosive properties of fluoride salts. Another extension would be more long-term simulations which would also run it with control rods inserted, to emulate the regular reactor operation.

Another would be running simulations for newer thorium fluoride salt reactors, such as the Copenhagen Atomics MSR, since the FUJI reactor resembles the MSRE with use of fluoride fuel salts and a corrosion resistant nickel alloy vessel (IAEA, 2016).

U-235 and U-238 have much higher delayed neutron fractions than Th-232 and Pu-239, (fraction of total fission neutrons emitted by unstable fission fragments) (Oka and Suzuki, 2013). Delayed neutrons are highly useful for maintaining a stable fission chain reaction without large temporary changes in reactor power. For use of the Th-232-Pu-239 mix, there would need to be further investigation into the role of delayed neutrons in reactor control, and the engineering and reactor operations required to accommodate this change (Oka and Suzuki, 2013).

Another extension to this extended essay would be to measure the time scale of neutron generation cycles, the time taken to reach certain k -effectives, to show the capability of increasing reactor power in a certain unit time. Another measurement that could have been used is the doubling time, which is the time taken to increase the quantity of fissile nuclides, or for neutrons, the time taken to double the total quantity of neutrons (European Nuclear Society, 2019).

Lastly, computations and measurements of reactor diagnostics, at the higher temperatures found in the fuel salt of a critical molten salt reactor, could be used for more extensively analysing the viability of this mix for reliable long-term commercial operation.

9. Conclusion:

In conclusion, referring to the research question, whether the thorium-plutonium mix is a viable substitute to enriched uranium, a lower concentration of fissile radionuclides in the beginning is required due to Pu-239 being a more potent fissile material. Additionally, Th-232 is a more potent fertile material in breeding Uranium-233. The connection between the concentration of fissile material and the k-effective, which measures the self-sustainability of the neutronics of a reactor, is noticeable and clear.

Enriched uranium should be appreciated functionally as a blend of fissile and fertile radionuclides, which provides opportunities to explore alternate, and more powerful blends of radioisotopes which can be used, such as a mixture of Th-232 and Pu-239 fluoride salts.

A fuel mixture consisting of Th-232 and Pu-239 has the unique advantage of being able to reach much higher k-effectives in reactors which use thermal neutrons, making it a stronger and more effective fuel for molten salt reactors, including in the MSRE.

10. References

Chidambaram, R, Ganguly, C. (1996). *Plutonium and thorium in the Indian nuclear programme. Current Science*. 70(1), 21–35.

Retrieved (September 1st, 2023) from: <http://www.jstor.org/stable/24097470>

Duderstadt, J, Hamilton, L. (1976). *Nuclear reactor analysis*.

Retrieved (September 1st, 2023) from:

http://milproj.dc.umich.edu/pdfs/books/1976_Nuclear%20Reactor%20Analysis.pdf

European Nuclear Society. (2019). *Doubling Time*.

Retrieved (May 27th, 2024) from: <https://www.euronuclear.org/glossary/doubling-time/>

Fratoni, Massimiliano, Shen, Dan, Ilas, Germina, and Powers, J. (2020). *Molten Salt Reactor Experiment Benchmark Evaluation*.

Retrieved (January 30th, 2024) from: <https://doi.org/10.2172/1617123>

IAEA. (2016). *Status Report - MSR-FUJI*.

Retrieved (June 13th, 2023) from: <https://aris.iaea.org/sites/MSR.html>

IAEA. (2014). *WIMSD-IAEA Lisa Meitner Library*.

Retrieved (April 18th, 2024) from: <https://www-nds.iaea.org/wimsd/xsplots.htm>

IEA. (2024). *Scaling up of innovative clean energy technologies needed to achieve net zero emissions targets globally - news*.

Retrieved (May 22nd, 2024) from: <https://www.iea.org/news/scaling-up-of-innovative-clean-energy-technologies-needed-to-achieve-net-zero-emissions-targets-globally>

Knudsen, E. (2022). *Notebook for running MSRE calculations directly from a CAD drawing*.

Retrieved (June 22nd, 2024) from:

https://github.com/openmsr/msre_docker/blob/main/MSRE.ipynb

Koelsch, T. (2016). *A Thesis Presented To The Department of Physical Sciences Emporia State University in Partial Fulfillment for the Degree Master of Science*.

Retrieved (January 30th, 2024) from:

<https://esirc.emporia.edu/bitstream/handle/123456789/3523/Todd%20Koelsch%20Spring%2016.pdf?sequence=1>

Oka, Y, and Suzuki, K. (2013). *Nuclear reactor kinetics and plant control*. Springer Japan.

ORNL. (1972). *Historic Molten Salt Reactor Experiment Brochure*.

Retrieved (September 1st, 2023) from:

[https://web.archive.org/web/20160303211133if_/https://dl.dropboxusercontent.com/u/15726934/Historic Molten Salt Reactor Experiment Brochure ORNL 1965-1972.pdf](https://web.archive.org/web/20160303211133if_/https://dl.dropboxusercontent.com/u/15726934/Historic_Molten_Salt_Reactor_Experiment_Brochure_ORNL_1965-1972.pdf)

Rehn, D. (2014). *Numerical Modeling of Core Disruption Accidents in Liquid Metal Breeder Reactors*.

Retrieved (December 22nd, 2023) from:

<http://large.stanford.edu/courses/2014/ph241/rehn2/>

Romano, P, K, Forget, B. (2013). “The OpenMC Monte Carlo Particle Transport Code.”.

Retrieved (March 23rd, 2024) from: *Annals of Nuclear Energy* 51 (January 2013): 274–281.
© 2012 Elsevier Ltd

Romano, P, K, Forget, B. (n.d.). *Quick Install Guide*

Retrieved (August 23rd, 2023) from:

<https://docs.openmc.org/en/stable/quickinstall.html>

Romano, P, K, Forget, B. (n.d.). *Official Data Libraries*

Retrieved (August 23rd, 2023) from: <https://openmc.org/official-data-libraries/>

Svalinn. (2023). DAGMC: Direct Accelerated Geometry Monte Carlo.

Retrieved (September 23rd, 2023) from: <https://github.com/svalinn/DAGMC>

Tucker, C. (2019). *How to Drive a Nuclear Reactor*.

Retrieved (May 27th, 2024) from: <https://link.springer.com/book/10.1007/978-3-030-33876-3>

Vlasov, A. (2023, September 14). *Thorium’s Long-Term Potential in Nuclear Energy*.

Retrieved (August 25th, 2024) from: <https://www.iaea.org/bulletin/thoriums-long-term-potential-in-nuclear-energy>

WNA. (2023). *Plutonium*.

Retrieved (February 2nd, 2024) from: <https://www.world-nuclear.org/information-library/nuclear-fuel-cycle/fuel-recycling/plutonium.aspx>

WNA. (2022). *Uranium Enrichment*.

Retrieved (October 22nd, 2023) from: <https://www.world-nuclear.org/information-library/nuclear-fuel-cycle/conversion-enrichment-and-fabrication/uranium-enrichment.aspx>

11. Appendix:

Raw Unprocessed Data for a Mix of U235-U238:

Concentration of U235 Relative to Total Uranium Concentration (Enrichment)	k-effective (No Units) $(0-1.5) \pm 1\%$
0.00	0.00139
0.05	0.353
0.10	0.562
0.15	0.695
0.20	0.799
0.25	0.890
0.30	0.960
0.33	0.998
0.35	1.06
0.40	1.09
0.45	1.13
0.50	1.14
0.55	1.16
0.60	1.20
0.65	1.22
0.70	1.24
0.75	1.27
0.80	1.30

0.85	1.31
0.90	1.33
0.95	1.35
1.00	1.36

Table 3: A table containing the raw unprocessed data for the relationship between enrichment and k-effective, when using a mix of U-235 and U-238 in the MSRE. All values are to three significant figures.

Raw Unprocessed Data for a Mix of Plutonium-239-Thorium-232:

Concentration of Fissile Plutonium-239 Relative to Total Radionuclide Concentration	k-effective (No Units) $(0-1.5) \pm 1\%$
0.00	0.000250
0.05	0.474
0.10	0.711
0.15	0.852
0.20	0.974
0.25	1.04
0.30	1.10
0.35	1.16
0.40	1.21
0.45	1.25
0.50	1.29
0.55	1.32
0.60	1.34
0.65	1.37

0.70	1.39
0.75	1.41
0.80	1.42
0.85	1.44
0.90	1.46
0.95	1.47
1.00	1.48

Table 4: A table containing the raw unprocessed data for the relationship between concentration of fissile nuclides and k -effective, when using a mix of Pu-239 and Th-232 in the MSRE. All values are to three significant figures.

The original simulation code, before the modifications made in the methodology can be seen below. To save space the detailed comments were also removed. This code and reactor model was taken from Eric Knudsen's OpenMSR library, (Knudsen, 2022), using DAGMC (Svalinn, 2023).

```
import openmc

graphite=openmc.Material(name='graphite')

graphite.add_element('C',1.0,'ao')

graphite.set_density('g/cc',2.26)

#Hastelloy N / INOR-8 nominal material composition from
#ORNL-TM-4189

inor=openmc.Material(name='inor')

inor.add_element('Ni',0.72)

inor.add_element('Mo',0.16)
```

```

inor.add_element('Cr',0.07)

inor.add_element('Fe',0.05)

inor.set_density('g/cc',9)

# LiF,BeF2,UF4,ZrF4 [0.67,0.23,0.05,0.0079] mol % @ 33% enrichment

molar_comp={'LiF':0.67,'BeF2':0.23, 'ZrF4':0.05, 'UF4':0.0079}

enrichment=0.33

salt=openmc.Material(name='salt')

salt.add_element('F',molar_comp['LiF']*1/2+molar_comp['BeF2']*2/3+molar_comp['ZrF4']*4/5
+molar_comp['UF4']*4/5,'ao')

salt.add_nuclide('Li7',molar_comp['LiF']*1/2,'ao')

salt.add_element('Be',molar_comp['BeF2']*1/3,'ao')

salt.add_element('Zr',molar_comp['ZrF4']*1/5,'ao')

salt.add_nuclide('U-235',enrichment*molar_comp['UF4']*1/5,'ao')

salt.add_nuclide('U-238',(1-enrichment)*molar_comp['UF4']*1/5,'ao')

salt.set_density('g/cc',2.2)

# The natural isotopes have been used for this alloy

# The density is set to that of Ni.

inconel=openmc.Material(name='inconel')

inconel.add_element('Ni',0.72,'ao')

inconel.add_element('Cr',0.20,'ao')

```

```
inconel.add_element('Fe',0.08,'ao')
```

```
inconel.set_density('g/cc',8.9)
```

```
helium=openmc.Material(name='helium')
```

```
helium.add_nuclide('He4',1.0,'ao')
```

```
helium.set_density('g/cc',1.0e-4)
```

```
materials=openmc.Materials([helium,salt,graphite,inconel,inor])
```

```
materials.export_to_xml()
```

```
materials
```

```
#geometry
```

```
h5m_filepath="../msre/msre_simple.h5m"
```

```
dag_univ = openmc.DAGMCUniverse(h5m_filepath)
```

```
geom = openmc.Geometry(root=dag_univ)
```

```
geom.export_to_xml()
```

```
xwidth = 350
```

```
yheight = 350
```

```
material_colors={salt:'red', inor:'lightblue', inconel:'blue',helium:'white',graphite:'gray'}
```

```
#xz plot
```

```
p1 = openmc.Plot()
```

```

p1.background='white'

p1.basis = 'xz'

p1.width = (xwidth,yheight)

p1.origin=(0,0,125)

p1.pixels = (800, 800)

p1.color_by = 'material'

p1.colors=material_colors

#xy plot

p2 = openmc.Plot()

p2.background='white'

p2.basis='xy'

p2.width=(xwidth, yheight)

p2.pixels = (800,800)

p2.origin=(0,0,100)

p2.color_by='material'

p2.colors=material_colors


plots=openmc.Plots([p1,p2])

openmc.plot_inline(plots)

# Create a neutron source for kick-starting

source_volume=openmc.stats.Box([-125,-125,0],[125,125,500], only_fissionable=True)

source = openmc.Source(space=source_volume)

```

```
source.angle=openmc.stats.Isotropic()
```

#Finally we build a settings object for OpenMC where we define parameters for the run.

```
settings = openmc.Settings()
```

```
settings.source = source
```

```
settings.batches = 20
```

```
settings.inactive = 5
```

```
settings.particles = 20000
```

```
settings.export_to_xml()
```

```
openmc.run()
```

```
mesh1=openmc.RegularMesh()
```

```
mesh1.dimension = [400,400,1]
```

```
mesh1.lower_left = [-125, -125, 90]
```

```
mesh1.upper_right = [125, 125, 110]
```

```
mesh1_filter = openmc.MeshFilter(mesh1)
```

```
mesh2=openmc.RegularMesh()
```

```
mesh2.dimension = [400,1,400]
```

```
mesh2.lower_left = [-175, -10, -50]
```

```
mesh2.upper_right = [175, 10, 400]
```

```
mesh2_filter = openmc.MeshFilter(mesh2)
```

```
t1 = openmc.Tally(name='flux1')

t1.filters =[mesh1_filter]

t1.scores = ['flux','fission']

t2 = openmc.Tally(name='flux2')

t2.filters =[mesh2_filter]

t2.scores = ['flux','fission']

tallies=openmc.Tallies([t1,t2])

tallies.export_to_xml()

import os

os.system("rm -f summary.h5 statepoint.20.h5")

openmc.run()

sp=openmc.StatePoint('statepoint.20.h5')

tl1=sp.get_tally(name='flux1')

tl2=sp.get_tally(name='flux2')

flux1=tl1.get_slice(scores=['flux'])

flux2=tl2.get_slice(scores=['flux'])

flux1.mean.shape=(400,400)

flux2.mean.shape=(400,400)
```

```

import matplotlib.pyplot as plt

import numpy as np

fig,(ax1,ax2)=plt.subplots(ncols=2,figsize=(20,16), constrained_layout=True)

ax1.set_xticks(np.arange(0,401,399/4))

ax1.set_xticklabels(np.arange(-175,176,350./4))

ax2.set_xticks(np.arange(0,400,399/4))

ax2.set_xticklabels(np.arange(-175,176,350./4))

ax1.set_yticks(np.arange(0,400,399/4))

ax1.set_yticklabels(np.arange(-175,176,350./4))

ax2.set_yticks(np.arange(0,400,399/4))

ax2.set_yticklabels(np.arange(-175,176,350./4))

ax1.set_xlabel('X / cm')

ax1.set_ylabel('Y / cm')

ax2.set_xlabel('X / cm')

ax2.set_ylabel('Z / cm')

im1=ax1.imshow(flux1.mean)

fig.colorbar(im1,ax=ax1,shrink=0.4)

im2=ax2.imshow(flux2.mean)

fig.colorbar(im2,ax=ax2,shrink=0.4)

#Similarly we can plot fission reactions:

```



```
fission1=t1.get_slice(scores=['fission'])
```

```
fission2=t2.get_slice(scores=['fission'])
```

```
fission1.mean.shape=(400,400)
```

```
fission2.mean.shape=(400,400)
```

```
fig,(ax1,ax2)=plt.subplots(ncols=2,figsize=(20,16), constrained_layout=True)
```

```
ax1.set_xticks(np.arange(0,401,399/4))
```

```
ax1.set_xticklabels(np.arange(-175,176,350./4))
```

```
ax2.set_xticks(np.arange(0,400,399/4))
```

```
ax2.set_xticklabels(np.arange(-175,176,350./4))
```

```
ax1.set_yticks(np.arange(0,400,399/4))
```

```
ax1.set_yticklabels(np.arange(-175,176,350./4))
```

```
ax2.set_yticks(np.arange(0,400,399/4))
```

```
ax2.set_yticklabels(np.arange(-175,176,350./4))
```

```
ax1.set_xlabel('X / cm')
```

```
ax1.set_ylabel('Y / cm')
```

```
ax2.set_xlabel('X / cm')
```

```
ax2.set_ylabel('Z / cm')
```

```

im1=ax1.imshow(fission1.mean)

fig.colorbar(im1,ax=ax1,shrink=0.4)

im2=ax2.imshow(fission2.mean)

fig.colorbar(im2,ax=ax2,shrink=0.4)

#define a logarithmic binning from 1keV to 10 MeV
ef=energyrange=openmc.EnergyFilter(np.logspace(3,7,200))

te = openmc.Tally(name='energy')

sf = openmc.MaterialFilter(salt)

te.filters =[ef,sf]

te.scores = ['flux','fission']

tallies.append(te)

tallies.export_to_xml()

os.system("rm -f summary.h5 statepoint.20.h5")

openmc.run()

sp=openmc.StatePoint('statepoint.20.h5')

tl=sp.get_tally(name='energy')

e1=tl.get_slice(scores=['flux'])

e2=tl.get_slice(scores=['fission'])

fig,ax=plt.subplots(figsize=(7,3.5),squeeze=True)

```

```

ax.plot(ef.values[:199],e1.mean[:,0,0])

ax.plot(ef.values[:199],e2.mean[:,0,0])

ax.set_xscale('log')

ax.set_yscale('log')

#geometry

os.system('rm -f materials.xml geometry.xml tallies.xml')

h5m_filepath="./msre/msre_control_in.h5m"

dag_univ = openmc.DAGMCUniverse(h5m_filepath)

geom = openmc.Geometry(root=dag_univ)

geom.export_to_xml()


b_w_conc={'Al2O3':0.3,'Gd2O3':0.7} # Wt. (D. Shen et.al. Nucl. Sc. & Eng., v. 195, pp. 825, 2021)

A_w={'Al':26.9815385,'Gd':157.25, 'O':15.99} #g/mol, (Webelements.com)

rho={'Al2O3':3.987,'Gd2O3':7.07} # g /cc, (Wikipedia: Aluminium_oxide & Gadolinium(III)_oxide)

b_mol_w={'Al2O3':2*A_w['Al']+ 3*A_w['O'],'Gd2O3':2*A_w['Gd']+3*A_w['O']}

bush_mol_comp={'Al2O3':(b_w_conc['Al2O3']/b_mol_w['Al2O3'])/(

b_w_conc['Al2O3']/b_mol_w['Al2O3'] + b_w_conc['Gd2O3']/b_mol_w['Gd2O3'] ),

'Gd2O3':(b_w_conc['Gd2O3']/b_mol_w['Gd2O3'])/(

b_w_conc['Al2O3']/b_mol_w['Al2O3'] + b_w_conc['Gd2O3']/b_mol_w['Gd2O3'] )}

bush=openmc.Material(name='bush')

bush.add_element('Al',2/5*bush_mol_comp['Al2O3'],'ao')

```

```

bush.add_element('Gd',2/5*bush_mol_comp['Gd2O3'],'ao')

bush.add_element('O',3/5*bush_mol_comp['Al2O3']+3/5*bush_mol_comp['Gd2O3'],'ao')

bush.set_density('g/cc',b_w_conc['Al2O3']*rho['Al2O3']+b_w_conc['Gd2O3']*rho['Gd2O3'])

```

```

t1 = openmc.Tally(name='flux1')

t1.filters =[mesh1_filter]

t1.scores = ['flux','fission','absorption']

```

```

t2 = openmc.Tally(name='flux2')

t2.filters =[mesh2_filter]

t2.scores = ['flux','fission','absorption']

```

```

tallies=openmc.Tallies([t1,t2])

tallies.export_to_xml()

```

```

materials=openmc.Materials([helium,salt,graphite,inconel,inor,bush])

materials.export_to_xml()

xwidth = 350

yheight = 350

material_colors={salt:'red', inor:'lightblue',

inconel:'blue',helium:'white',graphite:'gray',bush:'purple'}

```

#xz plot

p1 = openmc.Plot()

p1.background='white'

p1.basis = 'xz'

p1.width = (xwidth,yheight)

p1.origin=(0,0,125)

p1.pixels = (800, 800)

p1.color_by = 'material'

p1.colors=material_colors

#xy plot

p2 = openmc.Plot()

p2.background='white'

p2.basis='xy'

p2.width=(xwidth, yheight)

p2.pixels = (800,800)

p2.origin=(0,0,100)

p2.color_by='material'

p2.colors=material_colors

p3 = openmc.Plot()

p3.background='white'

p3.basis='xy'

```
p3.width=(xwidth/10, yheight/10)

p3.pixels = (800,800)

p3.origin=(0,0,100)

p3.color_by='material'

p3.colors=material_colors


plots=openmc.Plots([p1,p2,p3])

openmc.plot_inline(plots)

os.system("rm -f summary.h5 statepoint.20.h5")

openmc.run()

#Plot maps of the absorption

sp=openmc.StatePoint('statepoint.20.h5')


tl1=sp.get_tally(name='flux1')

tl2=sp.get_tally(name='flux2')


abs1=tl1.get_slice(scores=['absorption'])

abs2=tl2.get_slice(scores=['absorption'])


abs1.mean.shape=(400,400)

abs2.mean.shape=(400,400)
```

```
fig,(ax1,ax2)=plt.subplots(ncols=2,figsize=(20,16), constrained_layout=True)
```

```
ax1.set_xticks(np.arange(0,401,399/4))
```

```
ax1.set_xticklabels(np.arange(-175,176,350./4))
```

```
ax2.set_xticks(np.arange(0,400,399/4))
```

```
ax2.set_xticklabels(np.arange(-175,176,350./4))
```

```
ax1.set_yticks(np.arange(0,400,399/4))
```

```
ax1.set_yticklabels(np.arange(-175,176,350./4))
```

```
ax2.set_yticks(np.arange(0,400,399/4))
```

```
ax2.set_yticklabels(np.arange(-175,176,350./4))
```

```
ax1.set_xlabel('X / cm')
```

```
ax1.set_ylabel('Y / cm')
```

```
ax2.set_xlabel('X / cm')
```

```
ax2.set_ylabel('Z / cm')
```

```
im1=ax1.imshow(abs1.mean)
```

```
fig.colorbar(im1,ax=ax1,shrink=0.4)
```

```
im2=ax2.imshow(abs2.mean)
```

```
fig.colorbar(im2,ax=ax2,shrink=0.4)
```